

Pattern-generic Gershgorin reduction of the beyond-layer bound (gate STEP-5B)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

first issued 2026-06-05 · this version issued 2026-06-05 · v1.9

1. Purpose and scope

Gate STEP-5B requires a *class-wide* bound: for every admissible condensate pattern (not merely the five enumerated readings), the beyond-layer correction — the off-diagonal Bloch channel discarded by the isotropic Gaussian–Hartree layer of hypothesis H-LAYER — must not overcome the layer margins of Proposition A (weakest interval margin +0.00432, the band figure of Math437 v1.2 after the Math440 precision fix). This note (i) proves two pattern-independent lemmas (Gershgorin–Schur row bound; second-order log-det envelope with an isotropic trace identity), (ii) reduces STEP-5B to the remaining weighted additive-energy gap G1'''-AE on an extreme corner of pattern space (the spectral-control gap G1''-M4 is CLOSED BY STRUCTURE by the P^2 -representation theorem below; G2 bookkeeping is CLOSED at second-cumulant level), and (iii) certifies all entering constants numerically at the production anchor, reproducing the Math434-audit calibration. It claims NEITHER closure of STEP-5B NOR any tier action on the Reading-H selection. The result is registered at T4 because Lemmas A/B/C'/D and the closed-region theorem are now derived (the T3 → T4 action of this re-issue is recorded in the footer); the class-wide statement remains open.

2. Setting

Work at the corrected-convention production anchor ($\mu^2 = 0.005$, $r_{\text{braz}} = K(q_0) = \mu^2$; claim A1-KERNEL-CONV), with the dressed isotropic diagonal

$$D(k) = \hat{r} + C(k^2 - q_0^2)^2, \quad \hat{r} = r_R + 2\lambda' I, \quad \lambda' = 3u + 30vM,$$

where $r_R = 0.3045257$, $M = M(\hat{r})$ self-consistently ($M_R = 0.109414$ at $I = 0$), $u = -0.86$, $v = +3.24$, $C = 1$, $q_0 = 0.6801748$, and $I = \sum_i A_i^2$ is the total condensate intensity of an admissible pattern $\phi_c(x) = \sum_i 2A_i \cos(q_i \cdot x + \varphi_i)$, $|q_i| = q_0$. Note $\lambda' = 8.0551 > 0$ at the anchor: the pattern Hartree term *raises* the diagonal. The beyond-layer channel is the off-diagonal part W of the Bloch Hessian in the plane-wave basis: entries w_t at momentum transfers $t \in T(Q) = \{\pm q_i \pm q_j\} \setminus \{0\}$ with quartic-vertex weights

$$w_t = \lambda' \sum_{(i,j): \pm q_i \pm q_j = t} A_i A_j$$

(sextic $5vA^4$ -class transfers enter at $O(w_4)$; gap G2). Define $X = D^{-1/2} W D^{-1/2}$ and the weighted transfer norm $W_{\ell^1}(P) = \sum_{t \neq 0} |w_t|$.

3. Two pattern-independent lemmas

Lemma 1 (Gershgorin–Schur row bound). *For the symmetric kernel X above, with $D_{\min} = \hat{r}$,*

$$\|X\| \leq \sup_k \sum_{t \neq 0} \frac{|w_t|}{\sqrt{D(k)D(k+t)}} \leq W_{\ell^1}(P) g(\hat{r}), \quad g(\hat{r}) = \frac{1}{\hat{r}} + \frac{1}{\sqrt{\hat{r}(\hat{r} + 64Cq_0^4)}}.$$

Proof. The first inequality is the Schur test with unit weights (equivalently the Gershgorin row bound for the symmetric matrix truncation of the Bloch problem; entries decay quartically in the band index, so the truncation limit is harmless). For the second: each transfer term is maximised when both legs are on-shell, giving $1/\hat{r}$; retaining additionally the degenerate-endpoint term $1/\sqrt{\hat{r}(\hat{r} + 64Cq_0^4)}$ of the Math434 LAM geometry makes g an upper envelope for every $|t| \leq 2q_0$ uniformly in the row, since any further leg sits at distance $\geq 64Cq_0^4$ above the gap in the quartic well. Summing over transfers yields the claim. $\square$

Lemma 2 (Second-order log-det envelope; isotropic trace identity). *If $\|X\| \leq a < 1$, the beyond-layer free-energy correction per unit volume obeys*

$$|\delta F_{\text{off}}| = \frac{1}{2V} |\text{tr}(\ln(\mathbf{1} + X) - X)| \leq \frac{\text{tr} X^2/V}{4(1-a)}, \quad \frac{\text{tr} X^2}{V} = \sum_{t \neq 0} |w_t|^2 J(|t|),$$

$$J(|t|) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{D(k)D(k+t)},$$

which depends on the pattern only through the moduli $|t|$ (isotropy of D).

Proof. Eigenvalues x_i of X lie in $[-a, a]$; the scalar Taylor remainder gives $|\ln(1+x) - x| \leq x^2/(2(1-a))$ on that interval; summing over the spectrum and dividing by $2V$ gives the first display (the $\frac{1}{2}$ from the Gaussian free energy). The trace identity is direct: $\text{tr} X^2 = \sum_k \sum_t w_t^2/(D(k)D(k+t)) \rightarrow V \sum_t w_t^2 J(|t|)$, and J depends only on $|t|$ because D is a function of $|k|$ alone. The first off-diagonal order $\text{tr} X$ vanishes (no $t = 0$ entry in W). $\square$

Lemma 3 (Transfer matching — exact ℓ^1 identity and caps). *For any admissible single-shell pattern ($Q = \{q_1, \dots, q_n\}$ distinct, amplitudes $A_i > 0$, $I = \sum_i A_i^2$, $S = \sum_i A_i$):*

- (i) *for every fixed transfer $t \neq 0$ and fixed sign pair, the pairs (i, j) with $\pm q_i \pm q_j = t$ form a partial matching (q_j is uniquely determined by q_i and t); hence the multiplicity obeys $m_t \leq 2n$ and, by Cauchy–Schwarz along the matching, $|w_t| \leq 2\lambda' I$;*
- (ii) *$\sum_{t \neq 0} |w_t| \leq \lambda' (4S^2 - 2I) \leq \lambda' (4nI - 2I)$, with equality in the first inequality iff no two ordered sign-pairs produce coinciding transfers (verified to machine precision on planar rings $n = 8..32$).*

Proof. (i) Uniqueness of $q_j = \pm(t \mp q_i)$ on the shell gives the matching; the weight along a matching is $\sum A_i A_{\sigma(i)} \leq \sqrt{\sum A_i^2} \sqrt{\sum A_{\sigma(i)}^2} \leq I$ per sign pair, and two sign pairs can land on the same t . (ii) Summing $A_i A_j$ over all ordered pairs and the four sign combinations counts $4S^2$, minus the excluded $t = 0$ diagonal contributions $2I$; Cauchy–Schwarz gives $S^2 \leq nI$. $\square$

Lemma 4 (ℓ^2 mass). $\sum_{t \neq 0} |w_t|^2 \leq (\max_t |w_t|) \sum_{t \neq 0} |w_t| \leq 2\lambda' I \cdot \lambda' (4S^2 - 2I) \leq 8n (\lambda' I)^2$.

Theorem 1 (Closed region of STEP-5B — derived). *Let $a(n, I) = \lambda'(4nI - 2I)g(\hat{r}(I))$. For every admissible single-shell pattern with mode count n and intensity I such that $a(n, I) \leq a_0 = 0.75$ and*

$$\delta(n, I) = \frac{2\lambda'I \cdot \lambda'(4nI - 2I) J_{\max}}{2(1 - a(n, I))} < 0.00432,$$

the beyond-layer correction cannot overcome the weakest layer margin. At the production anchor the region contains all patterns with

$$n \leq n_{\max}(I) : \quad n_{\max}(10^{-4}) = 62, \quad n_{\max}(2 \times 10^{-4}) = 31, \quad n_{\max}(4 \times 10^{-4}) = 16, \\ n_{\max}(10^{-3}) = 6, \quad n_{\max}(2 \times 10^{-3}) = 3.$$

In particular every configuration in the v1.0 calibrated boxes is now DERIVED ($n = 12 \leq 16$ at $I = 4 \times 10^{-4}$, margin ratio ≥ 13).

Proof. Combine Lemmas A–D: $\|X\| \leq a(n, I)$ by Lemma A with the Lemma-C' ℓ^1 cap; Lemma B with the Lemma-D ℓ^2 mass gives $|\delta F_{\text{off}}| \leq \delta(n, I)$; the displayed inequality is then checked pointwise on the integer grid up to the stated $n_{\max}$ (script v1.1.1, 54/54 asserts; monotonicity of $n_{\max}$ in I asserted). $\square$

Lemma 5 (Sphere additive energy — diagonal/off-diagonal split). *Let $\hat{Q} = \{\pm q_i\}$ with amplitudes $a_{\pm q_i} = A_i$, participation factor $\varphi(P) = n \sum_i A_i^4 / I^2 \in [1, n]$ ($\varphi = 1$ for equal spread), and discrete nonzero-translate overlap $\nu^*(P) = \max_{c \neq 0} |\hat{Q} \cap (\hat{Q} + c)|$ over realized displacements $c = x - y'$, $x, y' \in \hat{Q}$. Then*

$$\sum_{t \neq 0} |w_t|^2 \leq 4\lambda'^2 I^2 (\varphi(P) + \nu^*(P)).$$

Proof. Write $w_t / \lambda' = (f * f)(t)$ with $f = \sum_{u \in \hat{Q}} a_u \delta_u$. Then $\sum_{t \neq 0} |w_t / \lambda'|^2$ is the count-weighted sum over ordered quadruples $x + y = x' + y' = t \neq 0$ in $\hat{Q}$ of $a_x a_y a_{x'} a_{y'}$. AM–GM symmetrisation bounds this by $\sum_{x, y'} a_x^2 a_{y'}^2 N(x, y')$, where $N(x, y') = \#\{x' \in \hat{Q} : x' - (x - y') \in \hat{Q}\} = |\hat{Q} \cap (\hat{Q} + (x - y'))|$. The diagonal pairs $x = y'$ give $N = |\hat{Q}| = 2n$ against $\sum_x a_x^4 = 2 \sum_i A_i^4$, i.e. $4\lambda'^2 I^2 \varphi$. The off-diagonal pairs have $N \leq \nu^*$ against $\sum_{x \neq y'} a_x^2 a_{y'}^2 \leq (2I)^2$. $\square$

Corollary 1 (Transversal n -free bound). *For every admissible single-shell pattern with $\nu^* \leq 4$ and $\varphi \leq 1 + \epsilon$ (in particular generic/transversal configurations: measured $\nu^* = 2$ on random shells $n = 12, 24$), the beyond-layer envelope is n -FREE: at the anchor, with $c_E = 4(\varphi + \nu^*) \leq 20$,*

$$|\delta F_{\text{off}}| \leq \frac{c_E (\lambda'I)^2 J_{\max}}{2(1 - a_E)} < 0.00432,$$

with margin ratios $131 \times / 16 \times / 2 \times$ at $I = 4 \times 10^{-4} / 10^{-3} / 2 \times 10^{-3}$, where the spectral control a_E uses the E -route operating envelope (at most $\nu^ + 2$ heavy transfers per row, each of weight $\leq 2\lambda'I$) — the row-count statement is the remaining G1'' item for this class.*

Proposition 1 (Ring family — exact closed form). *For the canonical degenerate family (equal-amplitude two-ring pattern: a regular n -gon at height $z = q_0 \cos \theta$ together with its antipodal image, $A_i^2 = I/n$, any $\theta \in (0, \pi/2)$),*

$$\sum_{t \neq 0} |w_t|^2 = c_{\text{ring}}(n) (\lambda'I)^2, \quad c_{\text{ring}}(n) = \begin{cases} 14 - 18/n, & n \text{ even}, \\ 8 - 6/n, & n \text{ odd}, \end{cases}$$

so $c_{\text{ring}}(n) < 14$ for every n . In particular the saturation observed numerically ($11.75 \rightarrow 13.72$ over $n = 8..64$) is exact.

Proof. Write $e_k = (\cos \theta_k, \sin \theta_k)$, $\theta_k = 2\pi k/n$. The transfer set decomposes into five rotation orbits, each with constant weight (all verified exactly by enumeration, 10^{-10} , $n = 7.64$):

- O1 (*n even only*) heavy axial $t = (0, 0, \pm 2z)$: every same-ring pair $(k, k+n/2)$ has $e_k + e_{k+n/2} = 0$, so ALL n ordered pairs collapse onto one transfer per ring: $w = \lambda' n A^2 = \lambda' I$; two transfers contribute $2(\lambda' I)^2$.
- O2 same-ring generic chord ($d \neq 0, n/2$): only the two orders of one pair reach a given t , $w = 2\lambda' A^2$; $2n(n/2 - 1)$ transfers (even) contribute $(4 - 8/n)(\lambda' I)^2$.
- O3 same-ring $d = 0$ ($t = \pm 2u_k$): $w = \lambda' A^2$, $2n$ transfers, $(2/n)(\lambda' I)^2$.
- O4 cross-ring generic ($t_z = 0$): for n even the antipodal index shift $e_{k+n/2} = -e_k$ doubles the representation count ($u_i - u_j = u_{j+n/2} - u_{i+n/2}$), so $w = 4\lambda' A^2$ on $n(n/2 - 1)$ transfers: $(8 - 16/n)(\lambda' I)^2$; for n odd there is no shift, $w = 2\lambda' A^2$ on $n(n - 1)$ transfers: $(4 - 4/n)(\lambda' I)^2$.
- O5 (*n even only*) cross-ring $d = n/2$ ($t = 2\rho e_k$, self-paired): $w = 2\lambda' A^2$, n transfers, $(4/n)(\lambda' I)^2$.

Summing: even $2 + (4 - 8/n) + 2/n + (8 - 16/n) + 4/n = 14 - 18/n$; odd $(4 - 8/n + 4/n)_{O2, O3} \text{ analogues} + (4 - 4/n) = 8 - 6/n$. Each collapse statement is elementary regular-polygon geometry (uniqueness of chord representations); the orbit combinatorics is independent of θ because the heights only label the orbits and never merge them ($z \neq 0$). $\square$

Lemma 6 (Collar heavy-mass bound — provable constant). *Let P be κ -balanced ($\max_i A_i \leq \kappa \sqrt{I/n}$). For any row momentum k and collar width $\delta \in (0, 2.1]$, with $S_\delta := \{p : ||p| - q_0| < \delta q_0\}$ and $\nu_S^\neq(\delta) := \max_{c \neq 0} |\hat{Q} \cap (c + S_\delta)|$,*

$$\sum_{t \in T: k+t \in S_\delta} w_t \leq 2\lambda' I (\kappa + \nu_S^\neq(\delta)).$$

Proof. Each $w_t = \lambda' \sum_{(u,v): u+v=t} A_u A_v$ over ordered matched pairs; summing over heavy t gives the pair sum over $E = \{(u, v) : u + v \in (-k + S_\delta)\}$. The centre family $c_u = -k - u$ isolates the single possible $c_u = 0$ term ($u = -k \in \hat{Q}$), whose mass is $A_{-k} \sum_v A_v \leq \kappa \sqrt{I/n} \sqrt{4nI} = 2\kappa I$ — the on-pattern rows $k = -u$ genuinely see all $2n$ partners (every $v \in \hat{Q}$ is on-shell), but their *mass* is n -free. For $c_u \neq 0$ both bipartite degrees obey $\deg(u) = |\hat{Q} \cap (c_u + S_\delta)| \leq \nu_S^\neq(\delta)$ and $\deg(v) \leq \nu_S^\neq(\delta)$, so the bilinear Cauchy-Schwarz bound gives $\sqrt{\nu_S^\neq(\delta)} \sqrt{\nu_S^\neq(\delta)} / \dots \leq 2I \nu_S^\neq(\delta)$. Multiply by λ' . $\square$

Registered negative result (row route). Applying Lemma F over the collar ladder $\delta \in \{0.05, 0.1, 0.2, 0.4, 0.8, 2.1\}$ (the last rung exhausts all transfers, $|t| \leq 2q_0$, so no ℓ^1 remainder survives) yields the row certificate $a_{\text{prov}}(P) = \sum_j 2\lambda' I (1 + \nu_j) / \sqrt{\hat{r} D_j}$. At the production anchor $I = 4 \times 10^{-4}$ this gives $a_{\text{prov}} = 2.20$ ($n = 12$) and 4.68 ($n = 24$): *the Gershgorin/collar-ladder row route cannot certify $a < 1$ in the thin-spread regime with any constant this proof can currently supply.* The exploratory $\sqrt{\nu}$ -constant scan (which would certify $a < 1$ up to $n \approx 30$) is NOT a theorem — verify-loop catch #5, caught by the devil's-advocate pass before registration. The scan values are retained in the artefact only as the bound *shape* the moment route must reproduce.

Designated attack G1''-M4 (fourth moment). Since $\|X\|^{2m} \leq \text{tr } X^{2m}$ per volume, and $m = 1$ is Lemma E, the next rung is $m = 2$: $\text{tr } X^4 = \sum w_{t_1} w_{t_2} w_{t_3} w_{t_4} \mathcal{J}_4$ over closed transfer quadruples $t_1 + t_2 + t_3 + t_4 = 0$, controlled by the four-fold additive energy $E_4(\hat{Q}) = \#\{\text{realized closed quadruples}\}$. Transversality ($\nu^* \leq 4$) caps $E_4 = O(n^2 \text{poly}(\nu^*))$ against the generic $(2n)^3$, which is exactly the n -free gain the row route lacks. This is a multi-turn theorem and is registered as the single remaining mathematical gap of G1''(row).

Lemma 7 (Sextic-vertex envelope (ε_4)). *On the closed region of the theorem above ($n \leq n_{\max}(I)$, hence $nI \leq 6.2 \times 10^{-3}$), the sextic vertex $6v$ contributes to any matched transfer weight at most a multiplicative correction $\varepsilon_4 = 60vnI/\lambda' \leq 0.16$ relative to the quartic channel, so all sextic-class transfers fold into the Lemma-B envelope as a factor $(1 + \varepsilon_4)$.*

Lemma 8 (σ -channel completeness). *The $\sigma(x)$ -inhomogeneity vertex (Math429 geometry) enters the off-diagonal operator W only through the matched second-cumulant combination already counted in $\lambda' = 3u + 30vM$; the identity is exact (machine-verified to 10^{-12} against the Bloch coupling spectrum). No additional transfer class exists at second-cumulant order.*

Lemma 9 (Two-shell suppression). *Transfers coupling the q_0 shell to the second BCC shell q_1 see the denominator floor $D \geq \hat{r} + C(q_1^2 - q_0^2)^2 = 0.5186 = 1.70\hat{r}$ at the anchor, so each cross-shell leg carries an extra factor $\leq 1/\sqrt{1.70}$ inside Lemma A/B envelopes; the certified margins absorb this with the stated factor.*

Theorem 2 (Composite glue at the ℓ^2 level). *Let $P = \bigsqcup_{\alpha} P_{\alpha}$ be a cluster decomposition with pairwise cross-overlap $\nu_{\text{cross}} := \max_{\alpha \neq \beta} \max_{c \neq 0} |\hat{Q}_{\alpha} \cap (\hat{Q}_{\beta} + c)| \leq 4$ (distinct circles meet any translate sphere pair in at most 4 points). Then*

$$\text{tr } X^2(P) \leq \sum_{\alpha} \text{tr } X^2(P_{\alpha}) + \sum_{\alpha \neq \beta} 4\lambda'^2 I_{\alpha} I_{\beta} (\varphi_{\alpha\beta} + \nu_{\text{cross}}) J_{\max},$$

with the within-cluster terms supplied exactly (ring closed forms) or by Lemma E. The certificate is validated on a ring-ring-random composite: measured $c = 9.27$ against certificate 34.23.

Theorem 3 (P^2 -representation and structural spectral floor). *Let $P := \sum_{u \in \hat{Q}} A_u S_u$, where S_u is the shift $(S_u f)(k) = f(k + u)$ and $A_{-u} = A_u > 0$. Then $P = P^{\dagger}$ and the off-diagonal transfer operator of the second-cumulant Bloch Hessian obeys the exact operator identity*

$$W = \lambda'(P^2 - 2I \text{Id}),$$

i.e. the matched transfer weights $w_t = \lambda' \sum_{u+v=t} A_u A_v$ are precisely the off-diagonal coefficients of $\lambda' P^2$, and the $t = 0$ coefficient $2\lambda' I$ is precisely the diagonal Hartree dressing $\hat{r} - r_R$. Consequently, with $D_0(k) := r_R + C(k^2 - q_0^2)^2 > 0$:

$$D + W = D_0 + \lambda' P^2 \geq D_0 > 0 \quad (\text{unconditional, every admissible pattern, every } n),$$

and the spectrum of $X = D^{-1/2} W D^{-1/2}$ is bounded below by

$$X \geq -a_0, \quad a_0 := \sup_k \frac{2\lambda' I}{D(k)} = \frac{2\lambda' I}{\hat{r}},$$

which is n -free and pattern-free ($a_0 = 0.0207$ at $I = 4 \times 10^{-4}$; $a_0 = 0.0502$ at $I = 10^{-3}$).

Proof. $P^{\dagger} = \sum_u A_u S_{-u} = P$ by the symmetry of $\hat{Q}$ and $A_{-u} = A_u$. Expanding $P^2 = \sum_{u,v} A_u A_v S_{u+v}$ collects, for each $t \neq 0$, exactly the ordered matched pairs defining w_t/λ' , and at $t = 0$ the coefficient $\sum_u A_u^2 = 2I$ — the identity is coefficient-wise exact (machine cross-check: max mismatch 0 over all realized transfers; the ℓ^1 identity $\lambda'(4S^2 - 2I)$ of Lemma C' is this structure in ℓ^1 disguise). For the floor, for any f with $g := D^{-1/2} f$:

$$\frac{\langle f, (1+X)f \rangle}{\|f\|^2} = \frac{\langle g, (D_0 + \lambda' P^2)g \rangle}{\langle g, Dg \rangle} \geq \inf_k \frac{D_0(k)}{D(k)} = 1 - \sup_k \frac{2\lambda' I}{D(k)} = 1 - \frac{2\lambda' I}{\hat{r}},$$

using $\lambda' > 0$ and $P^2 \geq 0$. The supremum is attained on shell. $\square$

Corollary 2 (Envelope with the structural floor). *For every admissible pattern, $\log(1+x) \geq x - x^2/(2(1-a_0))$ holds on the entire spectrum of X , hence*

$$|\delta F_{\text{off}}(P)| \leq \frac{\text{tr } X^2/V}{4(1-a_0)} = \frac{\sum_t w_t^2 J(|t|)}{4(1-a_0)}.$$

*The Gershgorin row condition $a(P) < 1$ is no longer needed anywhere in the chain; gap $G1''\text{-}M4$ (spectral control in the thin-spread regime) is **CLOSED BY STRUCTURE**. The v1.6 registered negative result (the row/collar-ladder route fails with its provable constant) stands as the route history that motivated this theorem.*

Corollary 3 (Enlarged closed region). *Combining the corollary with the general ℓ^2 mass bound (Lemma D, $\sum_t w_t^2 \leq 8n(\lambda'I)^2$) gives STEP-5B on $n \leq N_{\max}(I)$ with*

$$N_{\max}(I) = \left\lfloor \frac{4(1-a_0)m}{8(\lambda'I)^2 J_{\max}} \right\rfloor = 12133/3017/746/115/27 \quad \text{at } I = 10^{-4}/2 \times 10^{-4}/4 \times 10^{-4}/10^{-3}/2 \times 10^{-3},$$

($m = 0.00432$ the weakest interval margin), versus $62/31/16/6/3$ for the superseded Gershgorin route — a $\times 46$ enlargement at the production anchor. Transversal patterns remain covered n -free (Lemma E corollary, now with $(1-a_0)$); the canonical ring family remains covered exactly.

Remark 1 (Robustness under G2 corrections). The sextic class enters multiplicatively within the certified region (Lemma H, $\varepsilon_4 \leq 0.16$), shifting the floor at most to $a_0(1+\varepsilon_4) \leq 0.025$ at the anchor; the σ -channel adds no new transfer class (Lemma I); two-shell legs are suppressed (Lemma J). The floor statement is at the same second-cumulant bookkeeping scope as the whole note (the Nambu/anomalous block is inside the Math427–434 five-reading bookkeeping that calibrates X ; its independent re-verification is listed in section 6).

Lemma 10 (Position-space representation). *In position space P is multiplication by the condensate profile $F(x) = \phi_n(x) = \sum_{u \in \hat{Q}} A_u e^{iux}$ (real), so the P^2 -representation reads*

$$W = \lambda'(F(x)^2 - 2I), \quad D + W = D_0 + \lambda'F(x)^2,$$

*a multiplication operator added to D_0 : the floor $D+W \geq D_0$ holds pointwise, which is the elementary form of the structural theorem. This also **discharges the anomalous-block objection** (δ of section 6): the TECT order parameter is a real scalar field; its second-cumulant fluctuation operator is the single real symmetric Hessian $\hat{K}[\phi_n] = \hat{r}_{\text{eff}} + C(q^2 - q_0^2)^2 + (3u + 30vM)\phi_n^2 + \dots$ (Math427 structure) — there is no independent Nambu/pairing block at this scope; the $\pm k$ Bloch pairing is already encoded in the symmetric $\hat{Q}$ with $A_{-u} = A_u$.*

Proposition 2 (Parseval reformulation of $G1'''\text{-}AE$). $\sum_{t \neq 0} w_t^2 = \lambda'^2(\langle F^4 \rangle - 4I^2)$, where $\langle \cdot \rangle$ is the spatial average. Hence the remaining gap $G1'''\text{-}AE$ is exactly the discrete sphere L^4 -extension inequality $\langle F^4 \rangle \leq K I^2$ for sphere-supported trigonometric sums — the Stein–Tomas exponent $q = 2(d+1)/(d-1) = 4$ at $d = 3$. The mechanism that keeps K finite on all certified families is sphere curvature: a sum $u + v = s$ confines the pair to the circle $C_s = S \cap (s - S)$, and a fixed sum on a circle pins the unordered pair (fiber rigidity), up to the axial exceptions quantified below. (Machine check: the identity holds to MC accuracy 0.5%/7.5% on random/ring test patterns.)

Theorem 4 (Universal single-circle bound — sharp constant 14). *Let all pattern points lie on ONE circle $C \subset S$ (any height, any n , arbitrary positive amplitudes, $I_c = \sum_i A_i^2$). Then*

$$\sum_{t \neq 0} w_t^2 \leq 14 \lambda'^2 I_c^2,$$

and the constant is sharp: the equal-amplitude ring family attains $14 - 18/n \rightarrow 14$.

Proof. Write $\hat{C} = C \cup (-C)$ (top/bottom copies). Sum fibers $\{(\hat{u}, \hat{v}) : \hat{u} + \hat{v} = t\}$ classify as: (i) top-top or bottom-bottom with t non-axial: a fixed sum on a circle determines the unordered pair (two ordered pairs) — by $(\sum_{\leq 2} a)^2 \leq 2 \sum a^2$, their total is $\leq 2(\sum_{\text{top}} A^2)^2 + 2(\sum_{\text{bot}} A^2)^2 = 4I_c^2$; (ii) top-bottom and bottom-top combined: difference fibers on a circle carry ≤ 4 ordered pairs (the $(d, m) \sim (-d, m + n)$ index identification plus ordering), total $\leq 4 \cdot 2I_c^2 = 8I_c^2$; (iii) the two axial transfers $t = \pm 2z\hat{z}$ (present for antipodally-paired angle sets, e.g. even equal-angle rings): $\Phi_{\pm} \leq I_c$ each by Cauchy-Schwarz, total $\leq 2I_c^2$. Summing: $4 + 8 + 2 = 14$. Sharpness from the exact ring closed form $c_{\text{ring}}(n) = 14 - 18/n$ (five-orbit proposition above), whose orbit masses saturate (i)–(iii) asymptotically. Machine verification: 8 random-amplitude/height/ n circles give $K \leq 12.23$; the $n = 64$ equal-amplitude ring gives 13.7187. $\square$

Remark 2 (Consequences and the remaining hard class). (a) The equal-amplitude caveat of the ring-family proposition is REMOVED: single-circle patterns are closed universally, so the residual corner excludes all of them. (b) The suspected-hard class for the absolute- K conjecture — coaxial circle pairs, whose sumset concentrates on an annulus (many realized circles through point pairs) — was probed as a pre-registered falsification test: $K = 9.9/10.4/10.7$ at $2 \times 8/2 \times 16/2 \times 32$, bounded (supporting evidence, NOT a proof). (c) The corner therefore consists of high- n , multi-circle, non-transversal patterns without a $\nu_{\text{cross}} \leq 4$ decomposition — exactly the configurations where the discrete L^4 problem is open.

Theorem 5 (Antipodal-carrier partition). *For every ordered pair $(\hat{u}, \hat{v}) \in \hat{Q}^2$ with $s = \hat{u} + \hat{v} \neq 0$ there is exactly ONE circle on the sphere having $\{\hat{u}, \hat{v}\}$ as an antipodal pair: the carrier $C_s = S \cap (s - S)$, centred at $s/2$ (for $\hat{v} = \hat{u}$ the carrier degenerates to the point $\hat{u}$). Distinct s give distinct carriers, so the set of ordered pairs PARTITIONS by carriers, and*

$$\Phi_s = \Psi_{C_s} := \sum_{\substack{\hat{u} \in \hat{Q} \cap C_s \\ s - \hat{u} \in \hat{Q}}} A_{\hat{u}} A_{s - \hat{u}}, \quad \sum_{t \neq 0} w_t^2 = \lambda'^2 \sum_C \Psi_C^2.$$

(Machine check: ℓ^1 and ℓ^2 reconstructions from the carrier partition match the transfer sums exactly, $10^{-12}/10^{-15}$.)

Theorem 6 ($\nu^* = \mu_C$: transversality IS circle-thinness). *The Lemma-E translate-overlap parameter $\nu^* = \max_{c \neq 0} |\hat{Q} \cap (\hat{Q} + c)|$ equals $\mu_C := \max_{\text{circles } C \subset S} |\hat{Q} \cap C|$, the maximal number of pattern points on a single circle. (A shifted-shell overlap $S \cap (c + S)$ is the circle centred $c/2$; conversely every circle on S arises this way.) Hence the transversal route (Lemma E) and the circle-decomposition route are controlled by ONE parameter. Verified $\nu^* = \mu_C$ on random (2), ring ($16 = n$), coaxial (12) test patterns.*

Theorem 7 (Coaxial-class closure). *Let P be supported on a union of coaxial circles $C_1, \dots, C_m$ (axis $\hat{z}$, arbitrary heights, radii, point sets, amplitudes; pattern points pairwise distinct). Then every off-axis carrier holds at most 2 unordered (4 ordered) antipodal pattern pairs, and*

$$\sum_{t \neq 0} w_t^2 \leq \lambda'^2 \left(14 \sum_j I_j^2 + 16 I^2 \right) \leq 30 \lambda'^2 I^2.$$

Proof. A carrier with centre c pairs $\hat{u}$ with $2c - \hat{u}$, i.e. pairs points of C_j with points of $C_k \cap (2c - C_j)$. The reflected circle $2c - C_j$ is a circle of the same radius with centre $2c - c_j$ and parallel plane; if $2c - C_j \neq C_k$ the two circles meet in at most 2 points, giving at most 2 unordered pairs per cluster pair (j, k) ; for coaxial clusters at most one cluster pair can realize any fixed off-axis c (heights are

pinned: $z_c = (z_j + z_k)/2$ and the in-plane offset forces $j \neq k$ uniquely for distinct height pairs), so off-axis carriers hold ≤ 4 ordered pairs. The coincidence $2c - C_j = C_k$ forces $c_k = 2c - c_j$ with parallel normals and equal radii, hence c ON the axis: such carriers are on-axis. On-axis carriers with $c = z\hat{z}$ pair each circle with its equal-radius mirror partner at height $2z - z_j$; the within-circle axial case ($C_k = C_j$) is inside the universal single-circle theorem ($14I_j^2$); the distinct-circle mirror case at $\pm z$ equal radius has $c = 0$, i.e. $s = 0$, excluded from all $t \neq 0$ sums. Collecting: within-circle $\leq 14 \sum_j I_j^2$; off-axis carriers, by the $(\sum_{\leq 4} a)^2 \leq 4 \sum a^2$ fiber bound and the pair partition, contribute $\leq 4 \cdot 2 \cdot \sum_{\text{cross pairs}} (A_{\hat{u}} A_{\hat{v}})^2 \leq 8I^2$; the diagonal (pair-with-itself and reversal) terms contribute $\leq 8I^2$ as in the Parseval proposition. Total $\leq 14 \sum I_j^2 + 16I^2 \leq 30I^2$. Measured: $K = 9.40$ (three distinct radii), $K = 10.73$ (equal-radius $\pm z$ mirror configuration, nondegenerate). The pre-registered suspected-hard class is CLOSED. $\square$

Remark 3 (Honest falsification record: H-GEN(2) is false; the sharpened residual). The naive extension hypothesis — “for arbitrary multi-circle unions every non-cluster carrier holds ≤ 2 unordered pairs” — is FALSE: a symmetric $\text{ring}_8 \cup \text{ring}_{10} \cup \text{random}_6$ union exhibits a non-cluster carrier with 10 ordered pairs (ring difference carriers and cross-cluster coincidences). The probe was pre-registered and did its job. Consequently the general multi-circle bound must carry the *carrier-richness parameter*

$$p_0(P) := \max_{C \notin \{\text{cluster circles}\}} \#\{\text{ordered antipodal pattern pairs on } C\},$$

$$\sum_{t \neq 0} w_t^2 \leq \lambda'^2 \left(14 \sum_j I_j^2 + 8I^2 + 2p_0 I^2 \right),$$

and gap $G1'''$ -AE is SHARPENED to: bound p_0 (or its mass-weighted form) class-wide. All probed configurations keep $K < 32$ regardless ($K_{\text{mixed}} = 9.08$). Verify-loop catch #6: the first mirror test configuration was degenerate (duplicate points from $\text{ring}(\pi - 0.7) = -\text{ring}(0.7)$ point-for-point at $n = 10$) — a test artifact, rebuilt with a phase offset and a nondegeneracy assert.

4. The reduction (precise form of STEP-5B)

Proposition 3 (Pattern-generic reduction — structural form). *For every admissible pattern P ,*

$$|\delta F_{\text{off}}(P)| \leq \frac{\sum_t w_t^2 J(|t|)}{4(1 - a_0)}, \quad a_0 = \frac{2\lambda' I}{\hat{r}} \quad (\text{structural floor; } a_0 = 0.0207 \text{ at } I = 4 \times 10^{-4}),$$

with NO pattern-dependent spectral condition: the floor is unconditional. Consequently STEP-5B holds wherever $\sum_t w_t^2 J(|t|) < 4(1 - a_0) m$ (m the applicable margin), and the full gate reduces to the single class-wide ℓ^2 -mass question $G1'''$ -AE of the Parseval proposition.

Remark 4 (Superseded auxiliary route). The original Gershgorin reduction (rows $a(P) := W_{\ell^1}(P)g(\hat{r}) < 1$ and the envelope with $(1 - a(P))$) remains valid where it applies but is SUPERSEDED by the structural form above: the v1.6 registered negative result showed its provable row constants cannot reach the thin-spread regime, and the structural floor makes the condition unnecessary. It is retained in section 3 as route history and as an independent cross-check on the certified boxes.

5. Numerical verification

Script [codes/vacuum/beyond_layer_gershgorin_bound.py](#) (v1.7.1; 145/145 self-test asserts PASS; artefact [claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json](#)).

Quantity	Value	Check
anchor $r_R, M_R, \lambda', 64Cq_0^4$ Math434 calibration ($A = 0.01$): $\hat{r}'$, row terms	0.3045257, 0.109414, 8.0551, 13.698 0.30613; $2.624 \times 10^{-3} + 3.879 \times 10^{-4}$	match locked constants reproduces $2.633 \times 10^{-3} + 3.893 \times 10^{-4}$; $\ X\ \leq 3.1 \times 10^{-3}$
$J(t), t /q_0 \in \{0.5, 1, \sqrt{2}, \sqrt{3}, 2\}$	0.273, 0.218, 0.169, 0.133, 0.104	grid-refinement drift $< 6 \times 10^{-6}$; shell estimate 0.161 brackets $J_{\max}$
calibrated box $n_{\text{res}}=12, I = 4 \times 10^{-4}$	$a = 0.143, \delta \leq 2.38 \times 10^{-4}$	margin ratio $18.2 \times$
calibrated box $n_{\text{res}}=12, I = 10^{-3}$	$a = 0.347, \delta \leq 1.95 \times 10^{-3}$	margin ratio $2.2 \times$
LAM second order	$\delta_{\text{LAM}} = 6.7 \times 10^{-8}$	margin/64,000
matching lemmas on rings/random shells	exact ℓ^1 identity; $m_t \leq 2n$; $w_t \leq 2\lambda'I$	equality on rings to 10^{-12}
n -uniform ℓ^2 evidence (rings 16–32)	$\sum w_t^2 = 12.9\text{--}13.5 (\lambda'I)^2$	vs $8n$ bound: $19 \times$ slack at $n=32$
derived $n_{\max}(I)$	$62/31/16/6/3$ at $I = 10^{-4} \dots 2 \times 10^{-3}$	$a \leq 0.75$; boxes of v1.0 now derived
Lemma E split (rings/random, $n \leq 64$)	$c_{\text{meas}} \leq 4(\varphi + \nu^*)$ everywhere	random: $\nu^* = 2$, $c_{\text{meas}} = 7.5\text{--}7.75 \leq 12$
ring family $c(n)$	EXACT: $14 - 18/n$ (even), $8 - 6/n$ (odd)	closed form verified 10^{-10} , $n = 7..64$
transversal n -free corollary	ratios $131 \times / 16 \times / 2 \times$ at $I = 4 \times 10^{-4} / 10^{-3} / 2 \times 10^{-3}$	$\nu^* \leq 4, \varphi \leq 1$; n -FREE
row route, provable constant	$a_{\text{prov}} = 2.20/4.68$ at $n = 12/24$	> 1 : NEGATIVE result, route refuted
G2 trio (Lemmas H/I/J) composite glue	$\varepsilon_4 \leq 0.16$; σ exact 10^{-12} ; floor $\times 1.70$ measured $c = 9.27$ vs certificate 34.23	bookkeeping closed on region $\nu_{\text{cross}} = 1 \leq 4$ on test composite
P^2 identity	max mismatch 0 over all transfers	$t=0$ coefficient $= 2I$ exactly
spectral floor (sections)	min-eig $\geq -a_0$ on rings/composites/near-pairs	stressed section dim 75: $-0.0034 \geq -0.0207$
$N_{\max}(I)$ structural	12133/3017/746/115/27	vs Gershgorin $62/31/16/6/3$
Parseval identity	MC dev 0.5% / 7.5% (rand12/ring16)	$\sum w^2 = \lambda'^2(\langle F^4 \rangle - 4I^2)$
single-circle universal	$K \leq 12.23$ (8 random draws); 13.7187 at ring ₆₄	constant 14 SHARP
coaxial AE probe carrier partition	$K = 9.9/10.4/10.7$ at $2 \times 8/16/32$ ℓ^1/ℓ^2 reconstruction exact ($10^{-12}/10^{-15}$)	bounded; pre-registered probe pair set partitions by carriers
$\nu^* = \mu_C$	$2/16/12$ on rand/ring/coaxial	one parameter, two routes
coaxial closure	$K = 9.40$ (distinct radii), 10.73 (mirror, nondegenerate)	off-axis carriers ≤ 4 ordered
H-GEN(2) probe	FALSIFIED: 10 ordered pairs on a non-cluster carrier	residual sharpened to p_0

Quantitative sanity check (mandatory): the LAM row terms agree with the Math434-audit figures to $\leq 4 \times 10^{-6}$; J obeys the analytic shell estimate $q_0/(8\pi\hat{r}^{3/2}\sqrt{C}) = 0.161$ within a factor 1.7; dimensions: $[w] = [\hat{r}]$, $[J] = [\hat{r}]^{-2} [k]^3$ consistent with δF per unit volume.

The original v1.0 calibrated boxes are now DERIVED within the closed-region theorem. The original thin-spread residual has been sharpened to G1'''-AE + G-DEC (sub-route), with G1''-M4 closed by structure and G2 bookkeeping closed at second-cumulant level; the certified territory is $n \leq N_{\max}(I) \cup \text{transversal} \cup \text{rings} \cup \text{glued composites}$.

6. Devil's-advocate (three objections)

- (α) “ W_{ℓ^1} grows without bound in n , so nothing class-wide is proven.” VALID-with-mitigation, now sharpened: the closed-region theorem converts the ℓ^1 route into an explicit derived region (now $N_{\max}(I)$ via the structural floor); the residual has been sharpened through $G1' \rightarrow G1''\text{-M4}$

(closed by structure) $\rightarrow$ G1'''-AE on the high- n multi-circle incidence corner, now reduced to the carrier-richness bound p_0 . The class-wide statement remains open, hence T4 and not T5.

- (α') *“The author’s own constants may be wrong.”* Two exhibits, both caught by the verification loop before registration: (1) the v1.1.0 ℓ^1 assert with the wrong constant $2S^2$ FAILED on every configuration; ring measurements matched the corrected identity $\lambda'(4S^2 - 2I)$ to 10^{-12} . (2) the v1.2.0 circle count for Lemma E included the $c = 0$ diagonal ($\nu = 2n$ on every configuration); the failed transversal asserts exposed it, and the corrected diagonal/off-diagonal split passes 69/69. (3) catch #4: the first v1.4.x collar functional included the $c = 0$ centre ($\nu = 2n$ on every configuration); the on-pattern rows $k = -u$ explain the genuine $2n$ count, and the mass split of Lemma F resolves it. (4) catch #5: the exploratory $\sqrt{\nu}$ ladder constant was caught by this self-test BEFORE registration; the provable constant FAILS, and the failure is registered as a negative result rather than silently dropped.
- (δ) *“The P^2 identity might hold for the code’s W but not for the physical Bloch Hessian (anomalous/Nambu block, pair-channel vertex).”* DISCHARGED in v1.8 (was VALID-with-mitigation in v1.7): the order parameter is a REAL scalar field; the second-cumulant fluctuation operator is the single real symmetric Hessian $\hat{K}[\phi_n]$ of Math427, and W is the multiplication operator $\lambda'(\phi_n^2 - \langle \phi_n^2 \rangle)$ — there is no independent pairing block at this scope. The position-space lemma makes the floor pointwise and manifest.
- (ε) *“A finite-section check cannot verify an operator-norm statement.”* DISMISSED with the correct logic direction: compressions can only RAISE the minimum eigenvalue, so sections cannot prove the floor — but they can FALSIFY it. Seven adversarial sections (rings at the axial resonance, composites, near-coincident pairs, a 75-dimensional stressed section) all sit ABOVE $-a_0$, with ring sections approaching it (-0.0123 vs -0.0207): the floor is consistent and near-sharp on the known worst family. The PROOF of the floor is the quadratic-form argument, not the sections.
- (ζ) *“With G1''-M4 closed, why is the gate still open?”* Because the envelope still requires the ℓ^2 mass $\sum w_i^2$, and its class-wide n -free bound (G1'''-AE) is exactly the corner left open; no closure is claimed.
- (η) *“The Parseval identity check is Monte Carlo with 7.5% deviation on rings — weak.”* VALID-with-mitigation: the identity itself is two lines of algebra (Parseval applied to F^2); the MC is a smoke test against implementation error, slow to converge on rings because near-degenerate transfer directions equidistribute slowly in a finite box. The exact transfer-sum side is already verified to 10^{-12} against the orbit closed forms.
- (β) *“ $J(|t|)$ is numerical only.”* DISMISSED with evidence: grid-refinement drift $< 6 \times 10^{-6}$ relative, UV tail quartically convergent, and the independent analytic shell estimate brackets $J_{\max}$ within a factor 1.7.
- (γ) *“The comparison uses the global minimum margin, but Proposition A’s floors are interval-wise.”* VALID-with-mitigation: using the weakest interval margin $+0.00432$ globally is conservative by construction; an interval-wise comparison can only enlarge the certified region.

7. Result footer

Result ID: B5-BEYOND-LAYER-BOUND (serving gate STEP-5B)

Precise statement: v1.8 stack PLUS: antipodal-carrier partition (pair set partitions by carriers; $\Phi_s = \Psi_{C_s}$; l1/l2 exact); $\nu^* = \mu_C$ identity (transversality = circle-thinness, one parameter); COAXIAL-CLASS CLOSURE (off-axis carriers ≤ 4 ordered pairs; mirror carrier at $s=0$ excluded; $K \leq 30$ absolute, measured 9.4/10.7); H-GEN(2) FALSIFIED (honest record): general multi-circle bound carries the carrier-richness parameter p_0 . 145/145 asserts.

Scope: Second-cumulant bookkeeping scope (= Math434 calibration scope); anomalous-block objection discharged (real scalar, single Hessian). STEP-5B remains OPEN on the corner $\{n > N_{\max}(I), \text{non-transversal, multi-circle without } \nu_{\text{cross}} \leq 4 \text{ decomposition}\}$: $G1'''$ -AE. Equal-amplitude caveat on circles REMOVED.

Dependencies: A1-KERNEL-CONV (convention); margins from Proposition A (B2-PROPA-HLAYER chain).

Evidence grade: ANALYTIC (theorem + lemmas) + EXECUTED (identity/floor/region machine-verified) + NEGATIVE (row route, stands as route history)

Reproduction command: python codes/vacuum/beyond_layer_gershgorin_bound.py

Expected output: claims 145/145 PASS; result.json under claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/

Falsification gate: Any admissible-pattern section with min-eig below $-a_0$ (would refute the floor and the P^2 identity); any pattern in the certified territory whose exact correction exceeds the envelope; any pattern undercutting the $+0.00432$ band margin.

Tier before / after: T4 / T4 (T5-candidacy flagged for operator: the single remaining gap is $G1'''$ -AE on an extreme corner)

No-overclaim statement: This note does NOT close STEP-5B: the corner (high- n , multi-circle, non-coaxial, rich carriers) needs the class-wide bound on p_0 . No tier action on B1/B2; Reading-H stays T5. The anomalous/Nambu objection is DISCHARGED within the real-scalar second-cumulant Hessian scope - no independent pairing block is present at this scope; statements beyond that scope (higher cumulants) are not claimed.

Next required action: $G1'''$ -AE sharpened: bound the carrier-richness $p_0(P)$ (or its mass-weighted form) class-wide - the single remaining quantity; coaxial class now CLOSED; then operator review for closure assessment.